

Surgery for liver cancer

Surgery is the most effective treatment for primary liver cancer. There are two types of operation.

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Types of surgery for liver cancer

Surgery is the most effective treatment for primary liver cancer (also called hepatocellular carcinoma or HCC). The type of operation you have depends on:

- the number of tumours in the liver
- where they are in the liver
- how damaged the liver is from chronic liver disease.

There are two types of operation:

- a liver transplant – the surgeon removes the liver and replaces it with a liver from another person (a donor)
- a liver resection – the surgeon removes the part of the liver where the cancer is.

Only a few people with HCC can have surgery. Your surgeon and specialist nurse will tell you whether surgery is an option for you. They will explain what it involves and the possible complications or risks. An operation to your liver is major surgery. It is important to have all the information you need before you decide.

Getting Support

We understand that having treatment can be a difficult time for people. We are here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#).
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[Visit our liver cancer forum](#) to talk with people who have been affected by liver cancer, share your experience, and ask your questions

Liver transplant

The most common reason for a liver transplant is that you have liver disease that is too advanced for other treatments.

A liver transplant is only suitable for some people with HCC. It is not suitable if the cancer has grown into blood vessels or spread outside the liver. For example, it might be an option if you have:

- a tumour that is 5cm across or smaller
- up to 3 tumours and each is 3cm across or smaller.

You need to be quite fit to cope with a transplant. Your liver team will assess you very carefully to make sure it is a suitable treatment for you. Your liver surgeon also needs to be sure that the cancer cells have not spread.

It can take time, sometimes months, for a donor liver to become available. You may have other treatments during this time. These include:

- [tumour ablation](#)
- chemotherapy given directly into the liver called trans-arterial [chemoembolisation \(TACE\)](#)
- radiation using specialist techniques like [SIRT](#) or sometimes [SABR](#).

These treatments can help to control the cancer until you have your transplant. But for some people the cancer might progress. This means that a transplant is no longer possible. If you are no longer able to have a transplant, there are other treatments you can have.

Living donor liver transplant

It may be possible to have part of a liver transplanted from a living donor. This is a newer type of operation and is only available in a few hospitals. The donor has surgery to remove a part of their liver. This is transplanted into you after your liver is removed.

After a liver transplant

After a transplant, you need to take drugs called immunosuppressants for the rest of your life. These stop your body from rejecting the new liver. But they also make your immune system weaker. This means it may be less able to fight any cancer cells that might have spread.

Recovering from a liver transplant takes a long time. It will take time to build up your health and fitness. It can take many months before you get back to doing everyday activities.

We have not provided detailed information about your care before or after a liver transplant. Your liver team will talk to you about what to expect before and after the operation.

Liver resection

This operation is usually suitable for people who:

- have only 1 tumour or a limited number of tumours
- do not have cirrhosis or have early-stage cirrhosis.

[Cirrhosis](#) damages the liver and stops it from working properly. It is a risk factor for HCC.

The surgeon removes the part of the liver where the cancer is. The amount of liver they remove depends on the size and position of the tumour or tumours. They may remove:

- only a small part of the liver
- a whole lobe of the liver – this is called a hemi-hepatectomy.

The liver needs to be working well so that the remaining liver can cope after the operation. There are not usually any long-term side effects after a liver resection. This is because the remaining liver can grow bigger and work as it did before. This takes about 4 to 6 weeks. But you do not need to be in hospital while this happens.

Some people also have [tumour ablation](#) during surgery. This uses heat to destroy cancer cells.

Some people might have a treatment called [portal vein embolisation \(PVE\)](#) a few weeks before a liver resection. PVE redirects the blood flow to the healthy part the liver and encourages it to grow. You might have this when surgery is possible but there may not be enough remaining liver left afterwards to work properly.

Keyhole (laparoscopic) surgery

You might have open surgery for a liver resection. This is where the surgeon makes 1 large cut (incision). But surgeons often do a resection using keyhole surgery – this is also called laparoscopic surgery.

Instead of 1 large cut, the surgeon makes several small cuts. They insert a thin tube called a laparoscope into a cut to see into and work inside the tummy (abdomen). The laparoscope has a camera that gives a 3-dimensional (3D) magnified view of the inside of the body. The images are shown on a video screen.

Sometimes during keyhole surgery, the surgeon uses a fluorescent dye called indocyanine green (ICG). It shows up the blood vessels and the areas of the liver to be operated on. This helps the surgeon make sure the surgery is done very exactly. Because keyhole surgery leaves small wounds, you usually recover quicker.

Keyhole surgery may sometimes be done using robotic equipment. But this is not common. The surgeon controls instruments attached to robotic equipment using the video images as a guide.

Before your operation

If you smoke, try to [stop smoking](#) before your operation. This will help reduce your risk of problems, such as a chest infection. It will also help your wound heal after the operation. Your GP can give you more advice.

You will have tests to make sure you are well enough to cope with the operation. You usually have these a few days before your operation, at a pre-assessment clinic. They include tests on your heart and lungs.

At a clinic or when you are admitted to hospital, you will meet a member of the surgical team and a specialist nurse. They will talk to you about the operation. You may also meet the doctor who gives you the anaesthetic. They are called an anaesthetist.

Tell the surgical team if you have any questions or concerns about the operation. If you think you might need help when you go home, tell the nurses as soon as possible. This will give hospital staff time to help you make plans.

You will usually be admitted to hospital on the morning of your operation. You will be given special compression stockings (TED stockings) to wear during and after the operation. This is to prevent [blood clots](#) forming in your legs.

Preparing for surgery

Some hospitals have an enhanced recovery programme for certain types of surgery. Enhanced recovery programmes aim to reduce the time you spend in hospital and help you to recover as quickly as possible

For example, you may be given a diet plan to follow and exercises to do before surgery. You may also be given supplement drinks to take. This is to make sure you are as healthy as possible. After surgery, the nurses will get you out of bed and encourage you to start drinking and eating as soon as possible.

Getting fitter before an operation to avoid complications is called prehabilitation. It usually involves:

- stopping smoking, which reduces your risk of a chest infection
- stopping drinking alcohol, which helps your liver
- eating healthily
- being more physically active or exercising.

Your surgical team will tell you more about this and the [support available to help you](#).

After your operation

After a liver resection, you will be in intensive care or a high-dependency unit for about 24 hours. If you have a liver transplant, you will usually be in intensive care for a few days.

This is normal after major operations. There is a risk the liver may bleed after surgery. The doctors and nurses will monitor you carefully for bleeding. Your nurse will check your blood pressure regularly.

Moving around

The nurses will encourage you to start moving around as soon as possible. They will usually help you get out of bed the day after your operation or sooner. While you are in bed, it is important to move your legs regularly and do deep-breathing exercises. This helps prevent chest infections and blood clots. Your physiotherapist or nurse will show you how to do the exercises.

Drips and drains

For a short time after the operation, you may have some of the following:

- A drip going into a vein in your arm or neck. This is called an intravenous infusion. It gives you fluids until you can eat and drink again. It can also give you painkillers and other medications.
- 1 or more drainage tubes coming from your wound if you have had open surgery. These drain away extra fluids, such as blood. They are removed when the fluid has reduced.
- A fine tube that passes down your nose and into your stomach or small intestine. This is called a nasogastric tube. It drains fluids from your stomach so that you do not feel sick. You may need this for several days
- A small, flexible tube in your bladder. This is called a catheter. It drains urine (pee) into a bag. This means you will not have to get up to pass urine. You usually only have this for a couple of days.

Pain

There are effective ways to prevent and control pain after surgery. For the first few days, you are usually given painkillers through a pump. This gives you a constant dose of the painkillers.

You may have painkillers in the following ways:

- through a thin tube in your back (epidural)
- into a vein
- into the muscles close to your wound.

You may have a button you can press to give yourself an extra dose of painkillers if needed. This is called patient-controlled analgesia (PCA). It is set so that you cannot have too much painkiller.

When you no longer need painkillers through a pump, you can take them as tablets. Tell your nurses and doctors if you are in pain. They can give you the dose of painkillers that is right for you.

Your wound

For the first few days after surgery, you may have a dressing over your wound. The nurses will check your wound regularly to make sure it is healing well.

You usually have stitches that dissolve. But if you need to have any stitches or staples removed, these can be taken out after 10 days.

If you have had a liver transplant, the stitches or staples will be taken out after about 3 weeks. A nurse can do this in your own home or at your GP surgery.

Always tell your doctor if your wound becomes hot, painful or starts to leak fluid. These are possible [signs of infection](#). Contact the hospital straight away if this happens when you go home.

Going home

After a liver resection, most people can go home:

- 2 to 3 days after laparoscopic (keyhole) surgery for a smaller liver resection
- 4 to 5 days after keyhole surgery for a bigger liver resection
- 5 to 7 days after open surgery, if you have 1 large wound.

After a liver transplant, you will usually stay in hospital for 2 or 3 weeks.

Before you go home, you will be given painkillers to take for the next few weeks. Your nurse or pharmacist will talk to you about the tablets you need to take at home.

Recovering

It may take up to 3 months after a liver resection before you start getting back to normal. Recovery takes longer after a liver transplant. You will also need regular checks to make sure your body is not rejecting the new liver. Your doctor or nurse will tell you what to expect.

Your check-up will be at the outpatient clinic. Your doctor will ask you about your recovery and talk to you about the results of your operation. This is a good time to tell them about any problems you have had. Remember that you can contact your doctor or nurse before your appointment if you are unwell or worried about anything.

Some people take longer than others to recover. It depends on your situation, so do not put pressure on yourself. There are things you can do to help your recovery after surgery:

Avoid heavy lifting



For at least 8 weeks, you need to avoid lifting heavy loads, such as shopping or laundry. You will also need to avoid activities like vacuuming or gardening. This is to give your wound time to heal. Your specialist will tell you when you should be able to drive again.

Get enough rest and eat well



After any liver surgery, it is important to [get enough rest and eat well](#). This will help your recovery. We have more information about [lifestyle and wellbeing after treatment](#).

Do some gentle exercise



[Gentle exercise](#), such as regular short walks, will help build up your energy. You can gradually do more as you recover.

Do not drink alcohol



Your liver specialist will advise you not to drink alcohol for some time. How long this may be depends on your individual situation. This is to protect the liver while it is growing to replace the tissue that was removed.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our primary liver cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online



Available 8am to 7.45pm 7 days a week For confidential real-time support if you don't want to
Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help